B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:	I						
Paper Category:	DSC4-1 **(Major)**						
Paper Name:	Numerical Methods **(Group - II)** **Paper Code :** G06-0104						
Credit:	02			**Theory:**	2 Hrs./Week		
Marks:	**UA:**	30	**CA:**	20	**Total:**	50	

Course Objectives:

1. The course is designed to have a grasp of important concepts scientificallys in a scientific way.
2. The learner is expected to solve as many examples as possible to get complete clarity and understanding of the topics covered.

Course Outcomes:

At the end of this course, the student should be able to

1. Ability to appreciate real-world applications which use these concepts.
2. Skill to formulate a problem through Mathematical Modeling and programming.

Unit-I: Polynomial Interpolation Approximation and Errors:	**No. of Lectures:18**	**Weightage: 10-16 Marks**

Polynomial Interpolation Approximation : Argument and entries, equally spaced data and not equally spaced data, Finite difference operators: forward difference operator, backward difference operator, divided difference operator, Relation between these operators, Interpolation and Extrapolation, Newton's forward difference interpolation formula, Newton's backward difference interpolation formula, Lagrange's interpolation formula, problems, Divided Difference Operator , Divided difference table, Newton's divided difference interpolation formula (only formula without proof), problems.

Errors: Absolute error, relative error and percentage error, Normalized Floating point representation of real numbers, arithmetic operations on the numbers in normalized floating point notation: addition, subtraction, multiplication and division..

Unit-II: Matrices, Solution of System of linear equations And Solution of Non Linear Equations:	No. of Lectures:12	Weightage: 14-20 Marks

Matrices, Solution Of System of linear equations : Introduction, definition of a matrix and elementary results, types of matrices, operations on matrix, linear equations, system of linear equations, consistent system of linear equations, in consistent system of linear equations, ill and well-conditioned system of linear equations, solution of system of linear equations by using Jacobi's Iterative method and by using Gauss-Seidel iterative method, examples, characteristic polynomial, characteristic equation and characteristic roots of a matrix, Eigen values of a matrix(2x2 and 3x3), examples

Solution of non-linear equations: Introduction, definition of polynomial equation of degree n, linear equations, transcendental equations, non- linear equations. Location of roots, Algorithms to find root of given non-linear equations by using Bisection method, Regula Falsi Method, Newton Raphson Method. Solution of nonlinear equations by using bisection, regula-falsi and Newton Raphson method, examples.

Reference Books:

1. Introductory Methods of Numerical Analysis -- S.S. Sastry(Prentice Hall)
2. Computer Oriented Numerical Methods. -- Rajaraman
3. Introduction to applied Numerical Analysis. -- C. Richard, W. Hamming.
4. Numerical Methods for Science and Engineering -- R. G. Stanton (Prentice Hall)
5. Computers and Numerical Methods by E. Balguruswamy, (TMH).
6. Numerical Methods for Scientific and Engineering Computation --- M. K. Jain, S. R. K. Iyengar and R. K. Jain, New Age International Publisher.